Program : **Diploma in Information Technology**	
Course Code : **3265**	Course Title: **Computer Hardware and Networking Lab**
Semester : **3**	Credits: **1.5**
Course Category: **Program Core**	
Periods per week: **3 (L:0 T:0 P:3)**	Periods per semester: **45**

Course Objectives:

- To provide practical knowledge on basic Computer hardware installation and troubleshooting.
- To plan and implement a local area network and resource sharing using network components.

Course Prerequisites:

Topic	Course Code	Course Title	Semester
Basic IT Skills		Introduction to IT systems Lab	1

Course Outcomes

On completion of the course, students will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Assemble a P. C	10	Applying
CO2	Install O. S. and troubleshoot	9	Applying
CO3	Setup a simple LAN and do basic configuration	10	Applying
CO4	Setup a network using Simulator software	13	Applying
	Lab Exam	3	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7

CO1	3						
CO2	3			3			
CO3	3			3			
CO4	3			3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

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Module Outcomes	**Description**	**Duration (Hours)**	**Cognitive Level**
CO1	Assemble a P. C		
M 1.01	Identify components of a PC.	1	Applying
M 1.02	Identify ports and sockets of a PC	1	Applying
M 1.03	Experiment with SMPS	1	Applying
M 1.04	Experiment with Motherboard	1	Applying
M 1.05	Build a desktop PC	3	Applying
M 1.06	Identify the different beep sound produced by the motherboard	1	Applying
M 1.07	Identify the troubles in PC	2	Applying
CO2	Install O. S. and troubleshoot		
M 2.01	Experiment with Installation of Linux in a newly assembled PC	3	Applying
M 2.02	Experiment with hard disk partitioning and formatting	2	Applying
M 2.03	Experiment with hard disk file system	1	Applying
M 2.04	Experiment with Installation of application software in Linux	3	Applying
	Lab Exam – I	1.5	
CO3	Setup a simple LAN and do basic configuration		
M 3.01	Identify the structure of LAN and its components	1	Applying
M 3.02	Experiment with the Ethernet Switch and NIC card	1	Applying
M 3.03	Experiment with the colour code of LAN cable and Crimping	2	Applying
M 3.04	Build a simple LAN using Switch and PC	2	Applying
M 3.05	Utilize commands like ping, ifconfig etc.	1	Applying
M 3.06	Setup IP address and MAC address	1	Applying

M 3.07	Setup static and dynamic IP address settings	1	Applying
M 3.08	Experiment with DNS and Gateway settings	1	Applying
CO4	Setup a network using Simulator software		
M 4.01	Experiment with Network Simulation Software.	1	Applying
M 4.02	Experiment with basic router commands	2	Applying
M 4.03	Setup a network in a Simulator software with switches	2	Applying
M 4.04	Setup a Network in Simulator software with two LAN's and a router	2	Applying
M 4.05	Setup a Network in Simulator software with two LAN's and two routers	2	Applying
M 4.06	Open-ended experiment. **	4	Applying
	Lab Exam – II	1.5	

**** Suggested Open-ended Experiments:**

(Not for End Semester Examination but compulsory to be included in Continuous Internal Evaluation. Students can do open-ended experiments as a group of 2-3. There is no duplication in experiments between groups.)

1. Repair all the PCs that are not working in your campus.
2. Resolve all the connection problems in your campus network.

Text / Reference:

T/R	Book Title / Author
T1	Vikas Gupta, *Comdex Hardware and Networking Course Kit*, Dreamtech Press, 2018

Online Resources:

Sl.No	Website Link
1	https://www.tutorialspoint.com/computer_fundamentals/index.htm
2	https://www.javatpoint.com/hardware
3	http://www.codesandtutorials.com/hardware/